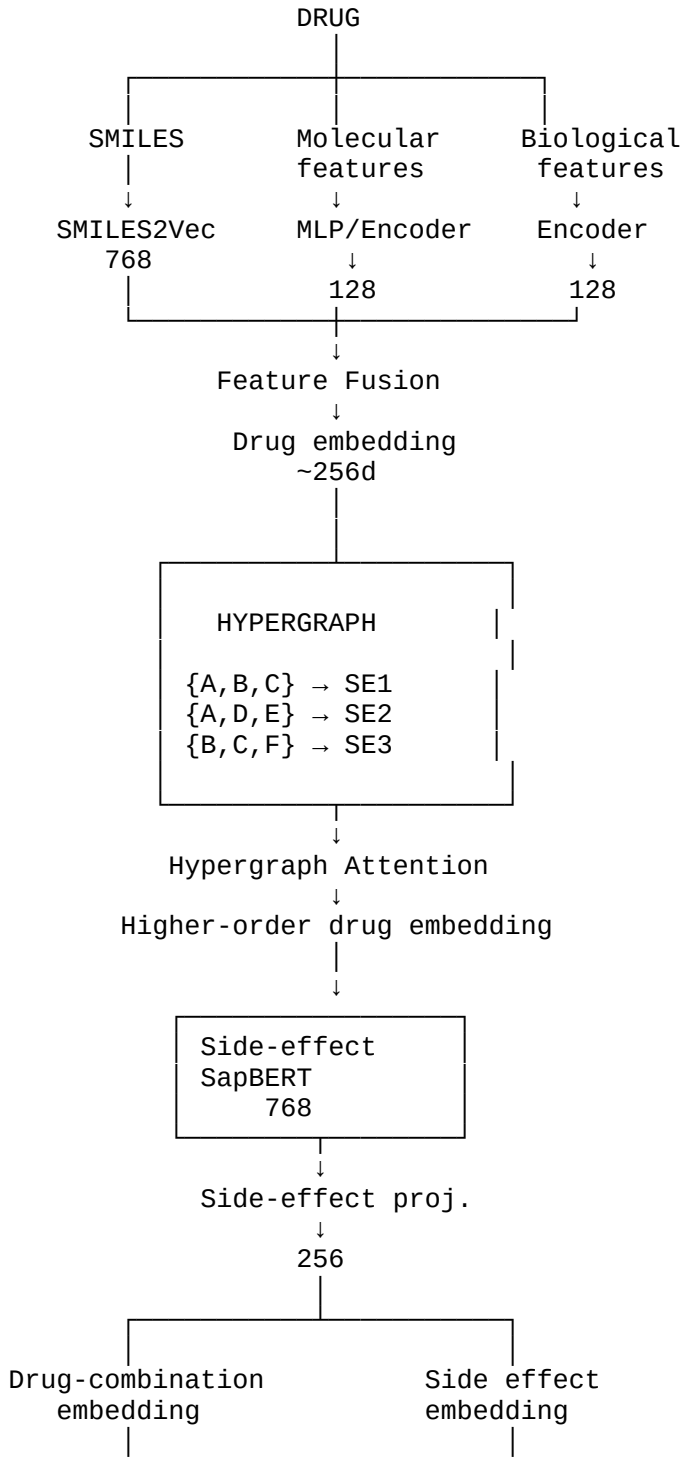
Right now, **do not start by designing a giant architecture**. Your first architecture should be a controlled extension of HODDI's HGNN-SA, because that gives you a fair baseline and a clear research contribution.

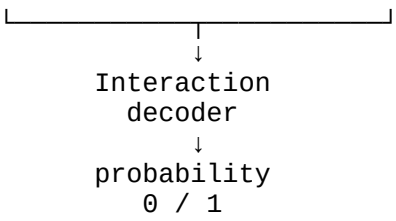
My recommendation: build HyperAttDDI in 3 layers

Core idea

\[\boxed{ \text{Multimodal Drug Encoder} + \text{Side-Effect Encoder} + \text{Higher-Order Hypergraph Attention} \rightarrow \text{Association Score} } \]

Your architecture should look roughly like this:





But **build it incrementally**.

Phase 1 — Your first model

Start with:

\[\boxed{\text{SMILES2Vec + SapBERT + Hypergraph}} \]

No extra features yet.

This gives you:

HyperAttDDI-v0

```
SMILES2Vec
    ↓
Drug embeddings
    ↓
Hypergraph Attention
    ↓
Drug-combination representation
    +
SapBERT side-effect representation
    ↓
MLP decoder
    ↓
P(y=1)
```

Why?

Because you need to answer:

> Does my architecture itself improve on HGNN-SA?

If you simultaneously add:

- molecular descriptors
- targets
- ATC
- fingerprints
- attention
- focal loss

and get 97% AUC, you won't know **why**.

Phase 2 — Make the attention actually meaningful

This is where I'd make your architecture different from a generic HGNN.

For a hyperedge:

$$ e=\{d_1,d_2,\ldots,d_k\} $$

you want the model to learn:

$$ \alpha_1,\alpha_2,\ldots,\alpha_k $$

where:

$$ \sum_i\alpha_i=1 $$

Then:

$$ h_e = \sum_i \alpha_i h_i $$

So instead of simply averaging:

$$ h_e = \frac{1}{k}\sum_i h_i $$

your model learns:

> Which drugs matter more within this particular combination?

For example:

```
{Warfarin, Aspirin, Drug C, Drug D}
              ↓
        Hyperedge attention
              ↓
Warfarin    0.45
Aspirin     0.38
Drug C      0.10
Drug D      0.07
```

That is a reasonable motivation for "HyperAttDDI."

Phase 3 — Make attention context-dependent

This is where I'd push the architecture further.

Don't calculate attention based only on the drug.

Calculate attention based on:

$$ \boxed{ drug + drug\ combination + side\ effect } $$

In other words:

> The important drug may depend on the adverse effect you're trying to predict.

For example, the same combination:

$$ \{A,B,C\} $$

could have:

```
Bleeding:
A → 0.60
B → 0.30
C → 0.10
```

but:

```
Kidney injury:
A → 0.15
B → 0.25
C → 0.60
```

That's much more interesting than generic attention.

Your attention equation could look like

For drug i in hyperedge e:

$$ q_e = W_q h_e $$$$ k_i = W_k h_i $$

and:

$$ \alpha_i = softmax \left(\frac{q_e^T k_i}{\sqrt d} \right) $$

Then:

$$ h_e = \sum_{i\in e} \alpha_i W_vh_i $$

But make the attention **side-effect-aware**:

$$ q_e = W_q[h_e \Vert s] $$

where:

- h_e = representation of the drug combination
- s = side-effect embedding
- $\Vert$ = concatenation

Now you're learning:

$$ \boxed{ \text{Which drug matters for this side effect?} } $$

That's a much stronger architectural story.

What about your additional features?

Don't put them in yet.

Once the basic architecture works, add them to the **drug encoder**.

I'd prioritize them in this order:

Level 1

Molecular structure

- SMILES2Vec
- Morgan/ECFP fingerprint
- Molecular descriptors

Level 2

Biological information

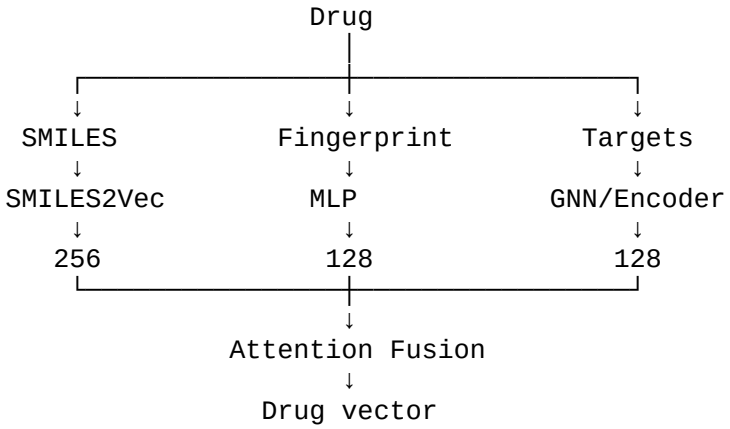
- Drug targets
- Enzymes
- Transporters
- Protein interactions

Level 3

Drug semantics

- Drug description
- Mechanism of action
- ATC classification

So eventually:



But this is **Phase 2 of your research**, not your starting point.

I would NOT use focal loss initially

Start with:

\[\boxed{\text{Binary Cross Entropy}} \]

Then later:

```
BCE
vs
Focal Loss
```

If focal loss helps, report it.

If it doesn't, remove it.

Otherwise you're adding complexity without knowing whether it contributes anything.

Your decoder should stay simple

Don't make your prediction head unnecessarily complicated.

After hypergraph attention:

\[h_{combo} \]

and side-effect embedding:

\[h_{SE} \]

then:

\[z = [h_{combo} \Vert h_{SE}] \]

and:

```
256 + 256
     ↓
   Linear
     ↓
   128
     ↓
   ReLU
     ↓
   Dropout
     ↓
   1
     ↓
  Sigmoid
```

Output:

\[P(y=1) \]

That's enough.

The architecture I would actually propose

Call it:

HyperAttDDI

Module 1 — Multimodal Drug Encoder

Initially:

$$ SMILES \rightarrow SMILES2Vec \rightarrow 256 $$

Later:

$$ SMILES2Vec + Fingerprint + Biological \rightarrow Fusion \rightarrow 256 $$

Module 2 — Higher-order Hypergraph Encoder

Given:

$$ e=\{d_1,d_2,\ldots,d_k\} $$

perform hypergraph message passing.

Module 3 — Side-effect-aware Hyperedge Attention

For each drug:

$$ \alpha_i = Attention(d_i,e,s) $$

Then:

$$ h_e=\sum_i\alpha_i h_i $$

This is the **core novelty** I'd focus on.

Module 4 — Side-effect Encoder

Initially:

$$ SapBERT \rightarrow 256 $$

Keep SapBERT frozen initially.

Module 5 — Interaction Decoder

$$ [h_e\Vert h_s] \rightarrow MLP \rightarrow P(y=1) $$

Your research experiments then become VERY clean

Experiment A

```
HODDI
 ↓
HGNN-SA
```

Published baseline.

Experiment B

```
HODDI
 ↓
Your HyperAttDDI
```

Same original features.

Question: Does your architecture help?

Experiment C

```
HyperAttDDI
+
Molecular features
```

Question: Do molecular features help?

Experiment D

```
HyperAttDDI
+
Biological features
```

Question: Do biological features help?

Experiment E

```
HyperAttDDI
+
Molecular
+
Biological
```

Question: Does multimodal fusion help?

Experiment F

Remove attention:

```
HyperAttDDI without attention
```

Compare against:

```
HyperAttDDI with attention
```

That's your **attention ablation**.

Experiment G

```
Mean pooling
vs
Attention pooling
```

This is particularly important.

You need to demonstrate that:

\[Attention > Mean \]

rather than simply assuming it.

And there's one experiment I REALLY want you to do

Evaluate performance as the number of drugs increases:

Interaction order	MLP	GAT	HGNN-SA	HyperAttDDI
2				
3				
4				
5				
6+				

This directly tests your central claim.

If your model is supposed to be **higher-order**, then I want to see:

\[\boxed{ \text{HyperAttDDI's advantage grows as } k \text{ grows} } \]

If it doesn't, that's an important scientific result too.

One thing I would change from your original proposal immediately

Don't frame your novelty as:

> **"We combine hypergraphs + attention + focal loss + additional features."**

That's just a pile of techniques.

Instead:

> **"We propose side-effect-conditioned attention over higher-order drug combinations, allowing the model to learn drug contributions that vary according to the adverse event being predicted."**

That's an actual **hypothesis**.

Then molecular/biological feature fusion becomes a secondary contribution.

So what should you do tomorrow?

Don't code HyperAttDDI yet.

Do these four things first:

1. Get HODDI running

Understand the exact data format.

2. Reproduce MLP

This should be your sanity check.

3. Reproduce HGNN-SA

This is your most important baseline.

4. Draw your proposed architecture

Before coding, we should specify exactly:

```
Input tensors
↓
Drug encoder
↓
Hyperedge construction
↓
Hypergraph message passing
↓
Side-effect-conditioned attention
↓
Combination representation
↓
Side-effect representation
↓
Decoder
↓
Binary prediction
```

Once those are fixed, **then** implement it.

And I would strongly recommend that we next design the architecture **mathematically and tensor-by-tensor** — e.g. "input shape = `[batch, max_drugs, 768]` → attention tensor = `[batch, max_drugs]` → hyperedge representation = `[batch, 256]`" — because that will expose any flaws in the architecture before you waste weeks coding it.